UltraBERT-JEV: Encoder models for System One Applications

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Abstract

A decision model picks one answer from a set of options in a single forward pass, without writing out its reasoning. It is cheap enough to run on every item in a large pipeline, for example to grade, filter or route text. This technical report gives a fine-tuning recipe for encoder-style decision models. We fine-tune UltraBERT-380M [32], an encoder pretrained with masked language modelling. We train it with a sigmoid-normalised loss: each option gets its own sigmoid score, which is divided by the sum over options, as in the expert gate of DeepSeek-V3 [14]. We compare it with sub-billion decoders. Qwen3-0.6B, fully fine-tuned on the same data with the LLM-as-Jev recipe [27], reaches **67.6%** test accuracy. The encoder reaches **72.1%**. It also scores higher on external benchmarks (**59.5%** against **56.1%**), and takes **29 ms** per decision against **94 ms** on an NVIDIA A100 80GB GPU. We also add an evidence head that marks, in the same pass, the sentences an annotator would mark as support for the answer. It is trained on human-marked evidence and on labels selected with a reward adapted from SelfCite [7].

1 Introduction

Software often has to make a decision among fixed options: route a support ticket, classify the intent of a message, or check a claim against evidence. A chat model can do this, but its answer has to be generated and parsed, which adds latency and formatting failures. Jev [2] is a decision model of this kind, and models built in its style are called Jev-style decision models. They return a probability distribution over the options without generating free-form text, so a new task needs only a prompt and a list of options [27]. LLM-as-Jev [27] shows that a decoder language model already works as such a decision model, with no change to its architecture, tokenizer or vocabulary. A frozen model handles any number of options, and fine-tuning with a loss over the option numbers and KL anchors helps compact models and tasks with many options. All of its models are causal decoders.

A decision is a prefill-only workload: the model reads the input once and produces a single token. Decoders are designed for the opposite case, long generation. Multi-query and grouped-query attention [1, 38] shrink the KV cache so that decoding can run with larger batches, and the parameters this removes are put back in the MLP layers. Du et al. [16] show what this costs when nothing is generated: the large intermediate tensors of the MLP layers cause periodic spikes in GPU memory during prefill, which limit the input length. An encoder has no generation step, so it does not keep a KV cache. It therefore does not need to shrink that cache and make up the parameters in larger MLP layers, which are the source of these memory spikes.

An encoder also reads the whole input in both directions. Its limitation was the head: a classifier head is fixed to one label set and must be trained again for each new task. A masked language model has no such limit. It already predicts a word at a masked position, so an option can be scored by asking whether the model fills a mask next to it with “yes” or “no”. Recent encoders such as ModernBERT [42] and UltraBERT [32] are pretrained with masked language modelling and can read long inputs, which decision tasks over documents need.

Reading in both directions matters for citing evidence as well. A decoder usually reads the document first and the question after it, so that the document’s KV cache can be reused across questions, as with prefix caching in vLLM [25] and RadixAttention in SGLang [50]. The document is then encoded without knowing the question, and its

representations cannot be used as they are to find the evidence for one question. An encoder reads the question and the document together, so every token is conditioned on the question.

We build an encoder decision model on this basis. It takes the same prompt and option list as LLM-as-Jev, puts one mask in front of each option, and scores all options in a single forward pass. It needs no new head, works with any number of options, and is small enough to run on every item of a pipeline. Our contributions are:

- **An encoder decision model.** We fine-tune UltraBERT-380M [32] with one mask per option and the "yes" versus "no" prediction at each mask as the option's score. It is trained on the same data as LLM-as-Jev.
- **A sigmoid-normalised loss.** Each option gets its own sigmoid score, divided by the sum over options, as in the gate of DeepSeek-V3 [14]. We compare it with the plain softmax over options and study how each loss treats the position and the label of the correct option.
- **A matched comparison with decoders.** We fully fine-tune Qwen3-0.6B [34] on the same data, tune the learning rate for every model, repeat the runs over seeds and evaluate on a held-out test split that was never used for a choice. We report accuracy, accuracy on external benchmarks, and time per decision, with two LoRA models [21] reproduced from LLM-as-Jev as references.
- **Evidence for the decision.** A decision model that generates no text gives no reasons, so we train an evidence head on the same encoder. Its labels come from human-marked evidence and from generated labels selected with a reward adapted from SelfCite [7]. We compare it with ContextCite [10] and AT2 [11] on agreement with human evidence over 200 rows, and on cost.

2 Method

2.1 How a row is presented

Figure 1 shows the same row in both formats, built from the training data described in Section 3.1. The options are shuffled each time a row is used, so the position of the correct option carries no information.

Decoder. The prompt follows LLM-as-Jev. The user message holds the state, the question, the options numbered [1] to [K], and one instruction: select one option and answer only with its number. The assistant turn is filled in up to "Best answer: [", and the model has to continue with the number of its choice and a closing bracket. Qwen3's thinking mode is turned off. Each option k is therefore the token sequence of "k]", and the options form a trie over these sequences.

Encoder. The state and the question are the same. The instruction asks to mark the correct option with "yes" and every other option with "no", and the options are listed as "[k] `[M]` description", with the tokenizer's mask token in front of each description. There is no chat template and nothing is generated. One forward pass gives a vector of logits at every mask, and the score of option k comes from the logits of the tokens "yes" and "no" at mask k, each taken with a leading space, as it appears after another word.

2.2 Loss functions

Decoder: trie loss with KL anchors. At each node of the trie the model chooses among the tokens that are legal there. The tree loss is the negative log-probability of the gold option, which is the product of the choices along its path, each normalised over the legal tokens only,

$$\mathscr{L}_{\text{tree}} = -\sum_{t} \log \frac{\exp z_t(g_t)}{\sum_{v \in \mathscr{V}_t} \exp z_t(v)}, \tag{1}$$

where z_t are the logits at step t, g_t the gold token and $\mathscr{V}_t$ the legal tokens. Three anchor terms keep the model close to its starting version p_0 [27]: $\mathscr{L}_{\text{mass}}$ keeps the share of probability on legal tokens, $\mathscr{L}_{\text{out}}$ keeps the distribution over illegal tokens, and $\mathscr{L}_{\text{pos}}$ keeps the next-token distribution at the start of the reply and after the closing bracket.

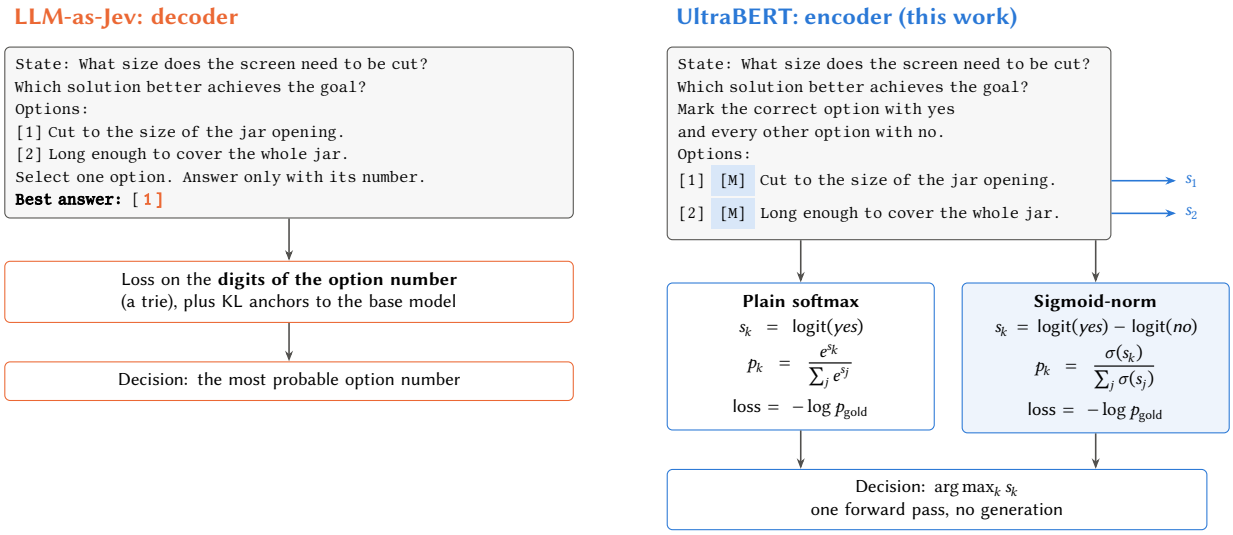


Figure 1. One row, two ways. The decoder reads the numbered options and writes the number of its answer; the loss is taken over the tokens of that number. The encoder gets a mask in front of each option and scores each option directly. Two ways to train it are compared: the plain softmax over the options, which scores an option by the "yes" logit alone, and the sigmoid-normalised loss, which scores it by "yes" minus "no". Each option's score goes through a sigmoid, and the sum in the denominator keeps the options competing.

They are KL divergences to p_0, computed over its 64 most likely tokens and one bucket for the rest. The loss is $\mathcal{L} = \mathcal{L}_{\text{tree}} + \lambda(\mathcal{L}_{\text{mass}} + \mathcal{L}_{\text{out}} + \mathcal{L}_{\text{pos}})$ with $\lambda = 1$.

Encoder: scores and normalisation. Let $\ell_k(v)$ be the logit of token v at mask k. We use two scores. The first is $s_k = \ell_k(yes)$. The second is $s_k = \ell_k(yes) - \ell_k(no)$, which also asks the model to reject the other options. Both give a decision by $\arg\max_k s_k$. The two losses differ in how the scores of the K options are turned into a distribution.

Plain softmax. The options compete through one softmax,

$$p_k = \frac{e^{s_k}}{\sum_{j=1}^{K} e^{s_j}}, \qquad \mathcal{L}_{\text{softmax}} = -\log p_g, \tag{2}$$

where g is the gold option.

Sigmoid-normalised loss. Each option's score first goes through a sigmoid, $\sigma(s_k)$, and these values are normalised by their sum,

$$p_k = \frac{\sigma(s_k)}{\sum_{j=1}^{K} \sigma(s_j)}, \qquad \mathcal{L}_{\text{sig-norm}} = -\log p_g. \tag{3}$$

This is the form of the gate of the DeepSeek-V3 mixture of experts [14], which scores every expert with a sigmoid and then normalises over the selected experts. Here the options play the role of the experts, normalised over all K options, with no selection and no balancing bias. Each $\sigma(s_k)$ lies between 0 and 1, and the options still compete through the sum. The gradient of the loss with respect to s_k is $(1 - \sigma(s_k))(p_k - \mathbf{1}[k = g])$, so an option that the model already accepts, with $\sigma(s_k)$ close to 1, adds little to the update. Since σ is increasing, the decision is the same $\arg\max_k s_k$. The sigmoid of a score can be read as that option's own probability of being correct, but nothing in the loss requires it: only the ratios enter p_k. Section 4.4 tests whether the trained model uses it this way.

3 Experimental setup

3.1 Training data

We follow LLM-as-Jev [27] and train on its Reason+Long mixture, with Nimble added. It has 13 datasets: intent routing (CLINC150 [26], MASSIVE [17], Bitext [6]), short reasoning (ReClor [45], LogiQA 2.0 [28], CosmosQA [22], PIQA [5], ANLI [29], ProofWriter [39]), long input (ContractNLI [23], MuSiQue [40], QuALITY [30]) and Nimble [4]. Each is turned into a state (the passages), a question and a list of options. We keep 1,000 training rows per dataset, 13,000 in total. Rows are filtered in this order:

1. **Duplicates.** Rows with the same input are collapsed to one, and rows with the same input but different answers are removed.
2. **Leakage between splits.** A training row is removed if its input, or the source document it comes from, also appears in the validation split.
3. **Benchmark overlap.** A training row is removed if it shares a 13-word n-gram, or an identical sentence of six or more words, with any evaluation benchmark.
4. **Length.** Prompts longer than 10,240 tokens, counted with the Qwen3.5-4B tokenizer, are dropped, never truncated.

The held-out rows are drawn from the validation split with the same duplicate and length filters, up to 500 per dataset and in whole source documents, giving 6,315 rows. We split them into a dev set of 1,300 rows, used to choose learning rates, and a test set of 5,015 rows that no choice depends on.

3.2 Models

Encoder. UltraBERT-380M [32] is fully fine-tuned with the format of Section 2.1, once with the plain softmax and once with the sigmoid-normalised loss.

Decoders. We fully fine-tune Qwen3-0.6B [34] with the loss of Section 2.2, with a frozen copy of the starting model as p_0. We also report two LoRA models [21], Qwen3-0.6B and Qwen3-4B, that we reproduced from the LLM-as-Jev paper on the same training set with the same loss. The encoder is fully fine-tuned, so the fully fine-tuned decoder is the matched comparison, and the LoRA models are references.

3.3 Implementation details

Every model is trained for one epoch with a batch of 32 rows. We sweep the learning rate with one seed per value and choose by accuracy on the dev set: $\{0.8, 1.5, 3, 5\} \times 10^{-5}$ for the encoder with the softmax, $\{0.8, 1.5, 2, 3\} \times 10^{-5}$ with the sigmoid-normalised loss, and $\{0.5, 1, 2\} \times 10^{-5}$ for the decoder. The two best Qwen rates differ by 0.1 points on dev, and we use 1×10^{-5}. At the chosen rate we repeat the run with different seeds, which change the order of the rows and the option shuffling, and report the average.

The metric is accuracy. ALL is the mean accuracy over the 13 datasets, so that every dataset counts equally. On the test set we average each row's correctness over the seeds of a model and report ALL of this average. The encoder is scored with fp32 weights under bf16 autocast, because bf16 weights change its long-document results (Section 5.4). The Qwen checkpoints are scored with bf16 weights.

3.4 External benchmarks and cost

To see whether training on the mixture changes behaviour outside it, we score six benchmarks as decision rows: BoolQ [8], MMLU [20], MMLU-Pro [41], ARC-Challenge [9], Winogrande [36] and SciQ [43], with 500, 1200, 800, 600, 500 and 400 rows. The sample sizes follow LLM-as-Jev, whose samples are not released, so ours are seeded uniform draws. External is the mean accuracy over the six. These benchmarks are not used for any choice.

Time per decision is the mean over the dev set on one A100, with one row at a time, as when serving single requests. The encoder is timed on the softmax checkpoint, which has the same architecture as the sigmoid-normalised one,

in the scoring precision. The decoders are timed with the reproduced LoRA checkpoints in bf16, with the LoRA weights merged, which have the same architecture as the fully fine-tuned model.

4 Results

4.1 Encoder against decoder

Table 1 and Figure 2 give the main comparison. With the same data and training budget, UltraBERT-380M with the sigmoid-normalised loss reaches 72.1% on the test set, against 67.6% for Qwen3-0.6B, a difference of +4.4 points. With the plain softmax the encoder reaches 69.7%, still +2.1 points above the decoder. The encoder also has fewer parameters, 380M against 0.6B. On the six external benchmarks the order is the same: 59.5% for the encoder and 56.1% for the decoder. Every test number is below its dev number by 0.6 to 2.0 points, because the learning rates were chosen on dev, and the ranking does not change. The reproduced Qwen3-4B LoRA model, at 77.3%, remains ahead of all sub-billion models, so our claims are limited to models below one billion parameters.

4.2 Training regime

The reproduced LoRA model of the same size reaches 63.0%, 4.6 points below the fully fine-tuned decoder. Both were trained on the same data, so this gap comes from updating all weights instead of a LoRA adapter, and accounts for about half of the 9.0-point gap between the encoder and the LoRA model. The external accuracy of fully fine-tuned Qwen3-0.6B (56.1%) is not above that of its LoRA model (56.9%); Section 5 shows how the learning rate and the number of epochs trade held-out against external accuracy.

4.3 Loss

The sigmoid-normalised loss is better than the plain softmax by +2.3 points on the test set, and by 1.7 points on the external benchmarks (59.5% against 57.8%). Its results vary less between seeds. The two losses differ in the score as well as in the normalisation; Section 5 separates the two and compares the labels that each loss predicts.

4.4 How strongly the options depend on each other

A softmax gate couples the experts of a mixture of experts, and DeepSeek-V3 replaces it with a sigmoid [14]. We test whether this matters for options. On the dev rows, we shuffle the options four times and remove one wrong option, and add a softmax model scored with "yes" minus "no" to separate the score from the normalisation (Table 2).

The answers become steadier from the score, not the sigmoid: "yes" minus "no" lowers the answer changes under shuffling from 15.6% to 12.7%, and the sigmoid adds nothing (13.7%). The sigmoid gives a small gain in label balance (total variation 0.139 against 0.219), calibration and dev accuracy. After training, the gold option's sigmoid averages only 0.049, so the model acts almost like a softmax.

4.5 Cost

The encoder takes 29 ms per decision on one A100, one row at a time, against 94 ms for Qwen3-0.6B and 170 ms for Qwen3-4B. Appendix A gives the batched throughput.

5 Analysis

This section describes what we saw when training UltraBERT. Unless noted, the numbers are from the dev set with one seed per setting.

5.1 Learning rate and training length

The learning rate that is best on dev, $1.5 \cdot 10^{-5}$ for both losses, is also among the best on the external benchmarks. Higher rates add little on dev and lower the external accuracy, down to 53.2% for the softmax at $5 \cdot 10^{-5}$. Qwen3-0.6B

Table 1. Matched comparison: average accuracy (%) over seeds. Test ALL is on the test set; Dev ALL and External are on the dev set and the six external benchmarks. Learning rates were chosen on dev. Bold: best below 1B parameters.

Model	Params	Dev ALL	Test ALL	External
UltraBERT-380M, sigmoid-norm	380M	**72.7**	**72.1**	**59.5**
UltraBERT-380M, softmax	380M	71.2	69.7	57.8
Qwen3-0.6B, full FT	0.6B	69.0	67.6	56.1
Qwen3-0.6B, LoRA (reproduced)	0.6B	64.5	63.0	56.9
Qwen3-4B, LoRA (reproduced)	4B	79.3	77.3	73.5

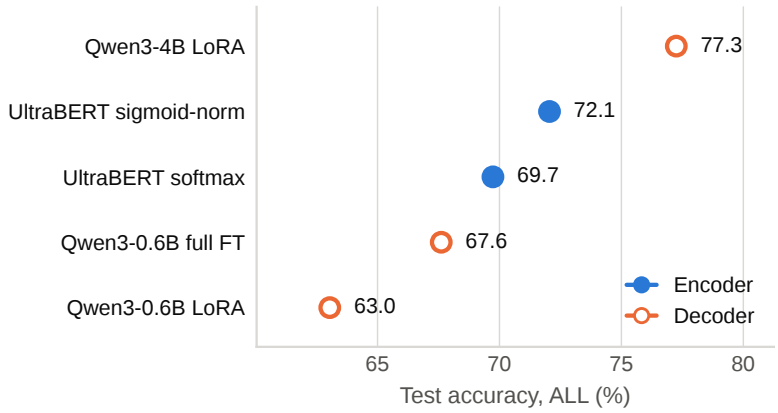


Figure 2. Test accuracy (ALL, the mean over 13 datasets, on the test set, never used for any choice), averaged over seeds; encoders filled, decoders hollow.

shows the same pattern more strongly: its external accuracy falls with every increase of the rate, from 57.4% at $5 \cdot 10^{-6}$ to 53.9% at $2 \cdot 10^{-5}$. A second epoch has the same effect. With the softmax at $3 \cdot 10^{-5}$ it raises dev accuracy from 70.7% to 73.9%, above any one-epoch encoder run, and lowers external accuracy from 56.2% to 55.1%. We report one epoch for all models, since a longer run fits the mixture at the cost of behaviour outside it.

5.2 Datasets

Table 6 in the appendix gives the test accuracy for every dataset, averaged over seeds. The sigmoid-normalised encoder is above the fully fine-tuned decoder on 10 of the 13 datasets. The gaps are largest on CosmosQA, ANLI and QuALITY, where it is ahead by 15.3, 9.9 and 9.8 points, and on MASSIVE (8.0) and MuSiQue (6.9). It is below the decoder on three datasets: Nimble (−6.6), LogiQA 2.0 (−2.2) and ProofWriter (−1.7). The 4B decoder is ahead of the encoder on all ten datasets outside intent routing. The gap is 12 to 19 points on ReClor, Nimble and LogiQA 2.0, and below one point on CosmosQA, ANLI and ContractNLI. On the three intent datasets, CLINC150, MASSIVE and Bitext, the encoder is level with or slightly above the 4B decoder, by 1.4 to 2.1 points.

5.3 Loss and score

Table 3 separates the changes between the plain softmax and the sigmoid-normalised loss. Scoring with the difference of the "yes" and "no" logits, with the softmax kept, raises the external accuracy by 2.0 points and the dev accuracy by 0.4 points. Replacing the softmax with the sigmoid-normalised loss then raises dev accuracy by a further 1.3 points, and MuSiQue most of all, from 64% to 76%. Nimble falls from 55% with the plain score to 43% with the sigmoid-normalised loss in this one run, on 100 dev rows. On the test set, averaged over seeds, the sigmoid-normalised model is instead 3.7 points above the softmax on Nimble. Replacing "yes" with "correct" did not help in this run: 70.6% against 71.1% with the "yes" logit, and 70.1% against 71.5% for the difference with "incorrect".

The two losses also predict different labels. On MuSiQue the correct answer is "Yes" in 48% of the rows. The sigmoid-normalised models predict "Yes" for 47 to 60% of the rows, against 64% and 69% for the two softmax variants (dev

Table 2. How strongly the options depend on each other, on the dev set (average scores). Each row is scored in four random option orders and once with one wrong option removed. TV: total variation between the predicted and the gold shares, over the three datasets with a fixed label set (label) and over option positions (position); lower is closer. Calibration error: expected calibration error of the top option's probability. $|s_k|$: size of the raw option scores. Bold: best in each row; accuracy is better higher, the other scores lower.

	Softmax, "yes"	Softmax, "yes" − "no"	Sigmoid-norm, "yes" − "no"
Dev accuracy (%)	71.4	71.5	**72.9**
Answer changes, options shuffled (%)	15.6	**12.7**	13.7
Answer changes, one wrong option removed (%)	5.9	**4.3**	4.9
Largest gradient norm in training	842	226	**190**
Label shares vs. gold (TV)	0.167	0.219	**0.139**
Position shares vs. gold (TV)	0.050	0.048	**0.041**
Calibration error, top option	0.038	0.030	**0.023**
Mean $\|s_k\|$	60.5	3.1	7.4

Table 3. Encoder variants on the dev rows, one seed (accuracy, %). The first row is the softmax with the "yes" logit, the setting used for the softmax results elsewhere. Bold: best in each column.

Variant	Dev ALL	External	MuSiQue	ContractNLI	Nimble
Softmax, "yes"	71.1	57.1	57.0	75.0	**55.0**
Softmax, "yes" − "no"	71.5	59.1	64.0	78.0	48.0
Sigmoid-norm, "yes" − "no"	**72.8**	**59.4**	**76.0**	**81.0**	43.0
Softmax, "correct"	70.6	56.8	48.0	77.0	52.0
Softmax, "correct" − "incorrect"	70.1	58.0	54.0	76.0	46.0

rows). On ProofWriter, where "Unknown" is correct for a third of the rows, the softmax never predicts it and the sigmoid-normalised models predict it for 2 to 12%. This class stays largely unlearned for every variant.

5.4 Numerical precision

The encoder has to be scored with fp32 weights under bf16 autocast. Its masked-token logits are of the order of 200, where bf16 can only step by 1 or 2, so options tie or swap. With bf16 weights the MuSiQue accuracy drifts by up to 8 points. The decoders are scored with bf16 weights, and re-scoring a Qwen checkpoint in another precision moved its dev accuracy by at most 0.8 points.

6 Citing the evidence

A decision model writes no text, so it gives no reasons. We add a small head on the frozen encoder that marks the document sentences supporting the chosen option, in the same forward pass. For each document token, it reads the token's hidden state and that of the chosen option's mask, and outputs the probability that the token is evidence. A sentence scores the highest probability among its tokens.

The head is trained on human-marked evidence where a dataset has it. For other rows, a labelling model proposes sets of sentences for the gold answer, and we keep the set with the best reward adapted from SelfCite [7] (Figure 3, Appendix B).

Comparison. We compare with the encoder's average attention, AT2 [11] and ContextCite [10] on 200 held-out rows with gold evidence, 50 each from ContractNLI, MuSiQue, Nimble and ProofWriter. This is not a comparison between equals: the head is trained on evidence labels, while AT2 and ContextCite are fitted to the model's response to ablations. We report sentence average precision (AP) against the gold sentences, how often the top sentence is

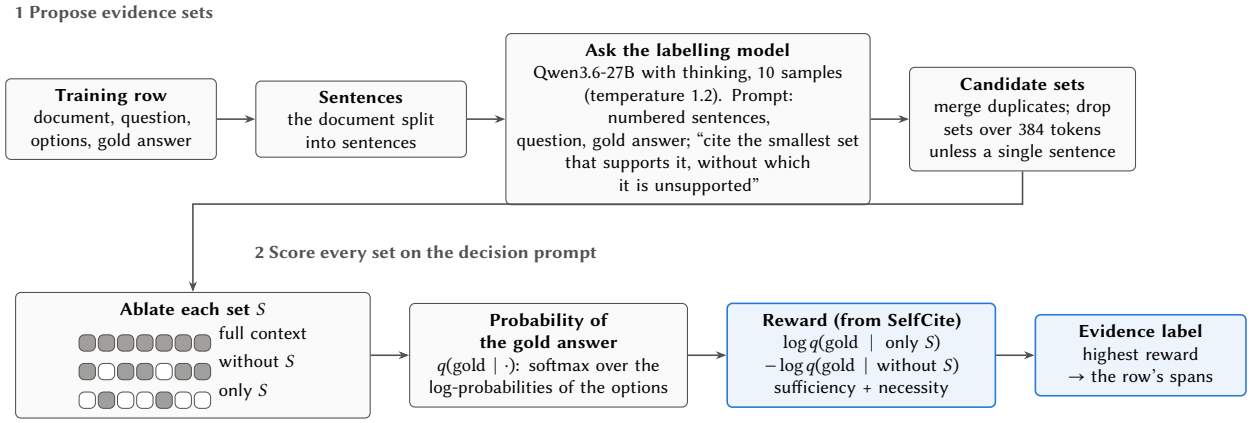


Figure 3. How evidence labels are generated for rows without human evidence. The labelling model, Qwen3.6-27B, proposes up to ten sets of sentences as evidence for the *gold* answer. The same model then answers the decision prompt three ways for each set S: with the full document, without S, and with only S: a good set is enough to keep the answer (sufficiency) and its removal loses the answer (necessity). The set with the highest reward becomes the label, which trains the evidence head. Unlike SelfCite, the answer scored is the gold answer and not the model's own response.

Table 4. Sentence evidence on 200 held-out rows with gold evidence, 50 per dataset. Sentence AP: average precision against the gold sentences, over all rows and per dataset (CNLI: ContractNLI, PW: ProofWriter). Top-1: share of rows where the top sentence is gold. LDS: how closely the scores follow the model; not computed for the heads. The first three methods and the first head use the encoder trained with the softmax. Each head is one training run. Bold: best in each column (lowest for time).

	Sentence AP							
Method	All	CNLI	MuSiQue	Nimble	PW	Top-1 (%)	LDS	ms per row
Average attention	0.26	0.08	0.19	0.40	0.37	15	0.11	**42**
AT2	0.33	0.34	0.29	0.30	0.39	22	0.18	47
ContextCite	0.41	0.54	0.34	0.33	0.41	45	**0.65**	1,704
Evidence head, softmax encoder	0.56	0.28	0.32	**0.97**	**0.68**	56	–	67
Evidence head, sigmoid-norm encoder	**0.68**	**0.56**	**0.50**	**0.97**	**0.68**	**74**	–	75

gold, and the time per row. For the baselines we also report LDS [10], the rank correlation between predicted and actual changes in the answer's probability when sentences are removed; we did not compute it for the head. AT2 and ContextCite ran on the softmax encoder, so Table 4 has a head for each encoder.

Results. On the same encoder, the head has a higher sentence AP than ContextCite and AT2 (0.56 against 0.41 and 0.33) in 67 ms per row, against 1,704 ms for ContextCite. The lead comes from Nimble and ProofWriter; on ContractNLI and MuSiQue, ContextCite is better. The head on the sigmoid-normalised encoder reaches 0.68, and is level with or above ContextCite on those two datasets. The words it marks have a precision of 0.33 and a recall of 0.35, so they are a rough guide, not an exact span.

Limits. The head predicts the evidence an annotator would mark, not what the model used; ContextCite has the highest LDS (0.65). On rows where the model is wrong, every method is scored against evidence for an option it did not choose.

7 Related work

Jev-style decision models. Most of the community decision models compared in LLM-as-Jev [27] use a decoder backbone and read the logits of an option letter or label, with the backbone frozen or tuned with LoRA. LLM-as-Jev studies this readout and when fine-tuning helps. AnyJev reads a typed decision from one prefill and corrects the label

prior and the effect of option order without training [46]. The Contrastive Language Model [24] embeds the state (the context and the question) and each candidate separately, with a frozen Qwen3-8B and two projection heads, and scores them by cosine similarity, so the candidate embeddings can be cached. Because the two are encoded apart, a candidate never attends to the context, and its training starts from tens of millions of pairs. All of these start from a decoder. We start from an encoder that reads the context, the question and all options together, and compare the two under the same training data.

Encoders that score options. Prompt-based classifiers use the masked-token head of an encoder with a verbalizer for a fixed set of labels [18, 37]. Two encoder decision models are listed on public Jev leaderboards [3]: Laya [12] and Von [31], both built on ModernBERT-large with one mask per option. Both add new layers on top of the encoder, and they are described in model cards. Laya trains its head with reinforcement learning on proper scoring rules, and Von uses a listwise softmax loss with a Brier term. We reuse the pretrained masked-token head and score each option by "yes" against "no", so there is no new head. We also compare two losses, match the decoders in data and training, report a held-out test split, and add evidence.

Option letters, position and label bias. Reading the letter of the chosen option requires a model to bind the letter to the option text [35]. Models favour certain option labels, and their accuracy changes when the options are reordered [33, 49]. Corrections are usually applied after training, by calibrating with a content-free input [48] or by dividing out a label prior and averaging over rotations of the options [46]. We shuffle the options during training and measure how the two losses differ in the labels they predict.

Encoders and decoders. Controlled studies pretrain encoders and decoders with the same recipe, and find that encoders do well on classification and retrieval [44] and that masked language modelling suits tasks that need text representations [19]. A much larger decoder can still beat an encoder on classification [47]. Decoder designs such as multi-query and grouped-query attention target long generation [1, 38], and Du et al. [16] study what they cost when nothing is generated (Section 1). Modern encoders keep long inputs and fast inference [32, 42].

Attribution and evidence. ContextCite learns a surrogate model from many ablations of the context [10]. AT2 learns to weight attention heads, and needs a decoder pass [11]. SelfCite fine-tunes a model to cite, using the change in the answer probability when the cited text is removed or kept alone as the reward [7]. Rationale benchmarks such as ERASER compare selected evidence with human rationales [15]. We train an evidence head on the encoder with labels from human-marked evidence and from the SelfCite reward, which gives the evidence in one pass.

8 Conclusion

We built a decision model on an encoder, UltraBERT-380M, with one mask in front of each option and the model's own "yes" versus "no" prediction as the score of the option. With the same data and full fine-tuning, it is more accurate than Qwen3-0.6B trained with LLM-as-Jev (72.1% against 67.6% on the test set), and takes 29 ms per decision against 94 ms. The sigmoid-normalised loss adds +2.3 points over the plain softmax and brings the predicted label rates closer to the true rates on MuSiQue and ProofWriter. An evidence head on the frozen encoder marks, in the same pass, the sentences an annotator would mark as support. On 200 rows from four datasets it ranks the annotated sentences higher on average than AT2 and ContextCite, which use no evidence labels, though ContextCite is better on two of the four datasets.

The results have limits. They are at the scale of 0.4 to 0.6 billion parameters, and a 4B decoder is still more accurate overall, except on the intent datasets. We used one encoder and one training mixture, in English, and compared against one decoder family. The evidence head predicts the evidence that an annotator would mark, and does not show what the model used; we did not compute its LDS, and ContextCite has the highest LDS of the methods we measured. We did not measure memory; that argument rests on Du et al. [16].

Two directions follow. The first is larger encoders, to see whether the advantage holds above one billion parameters. The second is to train the decision and the evidence together, so that the evidence head can also shape the encoder.

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A Implementation details

Table 5 lists the training settings. Both models train for one epoch, which is 406 steps at a batch of 32.

The encoder settings are those of the UltraBERT model card, and we change only the learning rate. No training row exceeds either token limit, because the data are cut at 10,240 tokens (Section 3.1).

Table 5. Training settings. Learning rates are in Section 3.3.

	UltraBERT-380M	Qwen3-0.6B (full)
Optimiser	AdamW, $\beta = (0.9, 0.98)$, $\epsilon = 10^{-6}$	AdamW
Weight decay	0.1	0.01
Schedule	5% warmup, linear decay to 0	100 warmup steps, linear decay to 10%
Gradient clipping	1.0	1.0
Precision	fp32 weights, bf16 autocast	fp32 weights, bf16 autocast
Attention	PyTorch SDPA	PyTorch SDPA
Gradient checkpointing	yes	yes
Token limit	32,768	16,384

Batched throughput. When the dev rows are scored in batches, the encoder processes 76.9 rows per second, against 16.4 for Qwen3-0.6B and 6.5 for Qwen3-4B. The encoder uses packed variable-length FlashAttention-2 [13] and the decoders use length-sorted padded batches, so part of this gap comes from the padding.

B Evidence head

We train one head on each fine-tuned encoder (sigmoid-normalised and softmax, both at a learning rate of $1.5 \cdot 10^{-5}$); the softmax one matches the encoder used by AT2 and ContextCite.

Model. For document token t and option k, the head reads $[h_t; h_k; h_t \odot h_k]$, the final hidden states of the token and of option k's mask. A two-layer perceptron (512 hidden units, GELU, dropout 0.1) maps it to one logit, whose sigmoid is the probability that the token is evidence for option k.

Labels. A token is positive if it overlaps an evidence span. We use gold evidence where it exists (Nimble, ContractNLI, MuSiQue, ProofWriter) and generated spans elsewhere (Figure 3): ANLI, QuALITY, LogiQA 2.0, CosmosQA and the remaining rows of ContractNLI, MuSiQue and ProofWriter. Qwen3.6-27B with thinking gets the numbered sentences, the question and the gold answer, and is asked for the smallest set of sentences that supports the answer and without which it is unsupported. We sample ten replies (temperature 1.2, top-p 0.9), merge identical sets, and drop sets over 384 tokens unless they are one sentence. The same model scores each set S on the decision prompt with the full context, without S and with only S ($1 + 2m$ prompts for m sets); $q(\text{gold} \mid \cdot)$ is the softmax over the options' summed token log-probabilities. The label is the set with the highest reward $\log q(\text{gold} \mid \text{only } S) - \log q(\text{gold} \mid \text{without } S)$, sufficiency plus necessity, with no size penalty since the token cap bounds the set. Unlike SelfCite, we score the gold answer, so the reward is informative only where the labelling model believes it. We did not check the generated labels against human evidence directly; the ablation below is an indirect check. There are 7,778 labelled training rows (7,390 for training, 388 for validation), with 5.2% positive tokens.

Training. Token-level binary cross-entropy, with positives weighted by the negative-to-positive ratio (18.1), on the frozen encoder. The head is conditioned on the gold option in training and on the chosen option at test time. AdamW (learning rate 10^{-3}, weight decay 0.01), 16 rows per step, three epochs; we keep the epoch with the best validation token average precision.

Evaluation. All methods are scored on the test rows with gold evidence and at least two sentences. A word is highlighted when its probability is at least a global threshold, tuned for word F1 on the dev rows with gold evidence: 0.78 for the sigmoid-normalised encoder and 0.73 for the softmax encoder. On the sigmoid-normalised encoder, the head's token average precision is 0.46 (0.17 for a random ranking), and 0.61 of its ten highest tokens are gold. On the softmax encoder, adding the generated labels to the gold ones raises sentence average precision from 0.52 to 0.56 and token average precision from 0.39 to 0.40.

C Additional results

Table 6 gives the test accuracy for each dataset, averaged over seeds.

Table 6. Test accuracy (%) per dataset, averaged over seeds. UB is UltraBERT-380M. Bold: best below 1B parameters.

Dataset	Rows	UB sig-norm	UB softmax	Qwen full	Qwen LoRA	Qwen-4B LoRA
CLINC150	400	**91.8**	**91.8**	86.3	79.8	89.8
MASSIVE	400	**84.9**	83.7	76.9	68.8	83.5
Bitext	400	**99.3**	**99.3**	96.0	94.8	97.2
ReClor	400	**59.9**	58.9	57.7	51.2	72.2
LogiQA 2.0	400	54.8	52.9	**57.0**	56.0	73.5
CosmosQA	400	**75.2**	74.8	59.9	55.0	76.0
PIQA	400	**75.1**	74.8	70.2	68.8	83.8
ANLI	400	**73.6**	73.1	63.7	61.0	74.0
ProofWriter	400	61.6	60.7	**63.3**	63.0	70.5
ContractNLI	393	**78.6**	73.8	76.1	65.6	79.1
MuSiQue	400	**69.8**	58.9	62.9	54.8	71.5
QuALITY	398	**61.3**	56.9	51.5	50.0	68.8
Nimble	224	50.7	47.0	**57.3**	50.9	64.3

D Examples

The examples show the decision of the sigmoid-normalised encoder and the token scores of its evidence head for one row of each of three datasets. The rows are the first of each dataset, in the evaluation order of the evidence study, that has at most 1,400 tokens; MuSiQue has none. Only the tokens that the head scores at or above its threshold of 0.78 for the chosen option are highlighted, darker for a higher probability, and tokens inside the gold evidence are underlined. The encoder chooses the correct option in all three rows; the rows were picked by the rule above and not by outcome.

ContractNLI. 1,376 tokens. The model chooses option 1, which is **correct**. Question: Statement: "Receiving Party may share some Confidential Information with some third-parties (including consultants, agents and professional advisors)." Does the contract entail the statement, contradict it, or not mention it?

k	Probability	Option	
1	49%	The contract entails the statement.	gold, chosen
2	32%	The contract contradicts the statement.	
3	19%	The contract does not mention it.	

Contract CONFIDENTIALITY & NON-DISCLOSURE AGREEMENT ATTACHMENT 3 - INVITATION FOR BID RFQ 205 CONSTRUCTION OF THE ASHBURN SHERIFF STATION THIS AGREEMENT is entered into by and between COUNTY OF LOUDOUN, VIRGINIA with offices at 1 Harrison St. S.E., Leesburg, Virginia 20175 (hereinafter "County") and with offices at (hereinafter "Bidder") on _____2015. Whereas, the County is seeking bids from Bidders to construct the Ashburn Sheriff Station (the "Project"); and Whereas, the Bidder desires to obtain architectural drawings that detail the design of said project for the sole purpose of compiling and submitting a response to Invitation for Bid RFQ 205 to construct the Project. NOW IT IS THEREFORE AGREED: 1. Purpose The purpose of this agreement is to allow the Bidder access to Confidential Information as defined herein detailing the design of the Project for the sole purpose of compiling and submitting a bid in response to the County's Invitation for Bid to construct the Project. This agreement is intended to ensure the safety and security of the Ashburn Sheriff Station. By signing this agreement the Bidder is stating that it will keep all Confidential Information (as defined in paragraph 4) in strict confidence and only use the Confidential Information on the Project identified herein. Further the Bidder warrants that its employees, partners, subcontractors, agents, and clients (collectively hereinafter the "Bidder") will be similarly bound by this agreement, or by other similar constraints of confidentiality. 2. Disclosure The Bidder shall not publish, or make and/or retain copies or disclose to any third party, any Confidential Information furnished to the Bidder except to such extent as may be necessary to carry out the Bidder's duties under this agreement and except to the extent that disclosure is required by judicial process in a legal proceeding, in which event Bidder will provide to the County notice of the request or demand to disclose Confidential Information. Within ten (10) days after award and/or completion of the IFB, or upon demand by the County, all Confidential Information, including all photographs, sketches, models, and drawings related to the design, layout and operation of said project shall be destroyed and files deleted from all of the Bidder's web sites and electronic devices (including but not limited to computers, cellular phones, etc.). Bidder shall also verify that all Confidential Information provided to partners, subcontractors, agents and clients has been destroyed. The successful Bidder shall destroy all Confidential Information within ten (10) days after final Project completion. 3. Indemnification Bidder agrees to indemnify and hold harmless the County, its elected officials, employees and agents from and against any and all liability, damages, claims, suits, liens, and judgments (including reasonable attorney's fees), of whatever nature, for injuries to or death of any person or persons, or loss of or damage to property, to the extent

attributable to the negligent acts of Bidder, its subcontractors or its respective agents, servants, or employees or such parties' related to said Confidential Information. 4. Confidentiality A. "Confidential Information" as outlined herein shall be used in strict confidence solely for the purpose allotted under this agreement and shall not be disclosed by the Bidder, its employees, partners, subcontractors, agents, or clients to a third party without the prior written consent of the County with the exception of the following; • It was already in the possession of the receiving party; • It is learned from or becomes publicly disclosed by a source with legal right to use and disclose same • It is identical to information which was developed by the receiving party independently of any Confidential Information received from the other party. B. Confidential Information means all information relating to the design, layout and operation of said Project. 5. Governing Law This agreement validity, construction and effect shall be governed under the substantive law of the Commonwealth of Virginia without the giving effect to any choice of law provision. Bidder expressly waives any objection to venue or jurisdiction of the Loudoun County Circuit Court, Loudoun County, Virginia. 6. Entire Agreement The governing terms and conditions of this agreement are expressly limited to the terms and conditions contained in this Agreement. This agreement constitutes the complete integration of all oral and written documents, is the entire and final agreement between the parties and may be amended only in writing signed by authorized officials of both parties. 7. Authority to Bind Firm in Agreement Person signing Agreement will show title or authority to bind the firm in an agreement. Name and authorized signature must appear below. Those authorized to sign are as follows: If a sole proprietorship, the owner may sign. If a general partnership, any general partner may sign. If a limited partnership, a general partner must sign. If a limited liability company, a "member" may sign or "manager" must sign if so specified by the articles or organization. If a regular corporation, the CEO, President or Vice-President must sign. Others may be granted authority to sign but the County requires that a corporate document authorizing him/her to sign be submitted with proposal. AGREED AND ACCEPTED BY: By: _____ Name (printed): _____ Title: _____ Date: _____ COMMONWEALTH OF VIRGINIA COUNTY OF _____ to-wit: Acknowledged before me this ____ day of _____, 2015. Notary Public Registration No.: My Commission Expires: _____

Nimble. 479 tokens. The model chooses option 2, which is **correct**. Question: Is the supplier bid ready to route for approval now under the stated rubric? Resolve pronouns using their nearest coherent antecedent.

k	Probability	Option	
1	46%	No — at least one stated requirement is unsatisfied now, so do not route for approval; route for supplier clarification.	
2	54%	Yes — every commercial, capacity, certification, technical, and current attachment requirement is satisfied, so route for approval.	gold, chosen

Statements Handoff coordinator: The PR-482 package was evaluated at the 2026-09-17 operational handoff using the quotation and documents then attached. Commercial reviewer: The supplier quoted a total price of $11,850 and a lead time of 28 calendar days for PR-482. Quotation reviewer: At the 2026-09-17 operational handoff, the supplier quotation for PR-482 stated a monthly gasket capacity of 640 gaskets. Request reviewer: At the 2026-09-17 operational handoff, PR-482 stated a minimum monthly gasket capacity of 600 gaskets. Certification reviewer: The supplier's ISO 9001 certification was current when the package was evaluated, and its current certificate was attached to the PR-482 package at that time. Document reviewer: The required material certificate was attached to the PR-482 package when evaluated. The submitted drawing bore the required signature and was also attached then. Technical reviewer: The gasket specification in the supplier quotation matches the technical specification in PR-482. Procurement specialist: Approval requires price no more than $12,000, lead time no more than 30 calendar days, capacity at least 600/month, current ISO 9001, a material certificate, and a signed drawing—all documents attached now. Technical specification owner: technical match does not waive the attachment requirements. Purchasing approver: Packages meeting every threshold, including boundary values, may be approved. If any required document is not attached now, route the package for supplier clarification.

ProofWriter. 195 tokens. The model chooses option 2, which is **correct**. Question: Statement: "Gary is not blue." Based only on the theory, is the statement true, false, or unknown?

k	Probability	Option	
1	1%	True: the theory implies the statement.	
2	69%	False: the theory implies that the statement is false.	gold, chosen
3	30%	Unknown: the theory implies neither the statement nor its negation.	

Theory Bob is big. Bob is blue. Bob is cold. Bob is white. Dave is cold. Dave is kind. Dave is nice. Gary is cold. Gary is kind. Harry is white. All white things are blue. All big, kind things are nice. If something is nice then it is big. If something is kind then it is nice. If something is rough then it is white. If something is big and nice then it is rough. All white, blue things are cold.